Multimedia Signal processing

Prof. Marco Tagliasacchi

July, 1 2014

1. **Question (8 pts)**:
   Consider a (somewhat simplified) document collection, which is described by the following term-document matrix

$$A = \begin{bmatrix} 10 & 8 & 2 & 8 & 0 & 2 \\ 0 & 1 & 4 & 3 & 2 & 0 \\ 1 & 3 & 0 & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}, \tag{1}$$

   where $A_{i,j}$ represents the number of times term i (row index) appears in document j (column index).
   (a) Compute the number of bits needed to Huffman code each document (assigning one codeword to each term).
   (b) Compute the number of bits needed to arithmetic code each document.

   Solution Let X denote a discrete source, whose symbols are the four possible terms included in the document collection, namely t_1, t_2, t_3, t_4.

   Let m_i denote the number of times that a term i appears in the collection. Then, $m_1 = 30$, $m_2 = 10$, $m_3 = 10$, $m_4 = 2$.

   Let p_i denote the probability that term i appears in a document. Then, $p_1 = 30/52$, $p_2 = 10/52$, $p_3 = 10/52$ and $p_4 = 2/52$

   (a) Assuming a coding scheme based on Huffman coding, we find the following codewords

symbol	probability	codeword	length
t_1	30/52	0	1
t_2	10/52	00	2
t_3	10/52	011	3
t_4	2/52	010	3

   Encoding the documents would require a number of bits equal to

$$R_1 = 1 \cdot 10 + 3 \cdot 1 = 13 \tag{2}$$
$$R_2 = 1 \cdot 8 + 2 \cdot 1 + 3 \cdot 3 = 19 \tag{3}$$
$$R_3 = 1 \cdot 2 + 2 \cdot 4 = 10 \tag{4}$$
$$R_4 = 1 \cdot 8 + 2 \cdot 3 + 3 \cdot 1 = 17 \tag{5}$$
$$R_5 = 2 \cdot 2 + 3 \cdot 4 + 3 \cdot 1 = 19 \tag{6}$$
$$R_6 = 1 \cdot 2 + 3 \cdot 1 + 3 \cdot 1 = 8 \tag{7}$$
$$\tag{8}$$

(b) Assuming a coding scheme based on arithmetic coding, we can approximate the number of bits to encode each document with $l = \lceil -\log_2 p(\mathbf{x}) + 1 \rceil$,

symbol	probability
t_1	30/52
t_2	10/52
t_3	10/52
t_4	2/52

Encoding the documents would require a number of bits equal to

$$R_1 = \lceil \log_2 [(30/52)^{10} \cdot (10/52)^1] \rceil + 1 = 12 \tag{9}$$

$$\dots \tag{10}$$

2. **Question (5 pts)**
   (a) Illustrate the principles of transform coding.
   (b) Describe how the bitrate is allocated to each transform coefficient.

   Solution

   See slide set `MMSP_1.4_Transform coding`.

3. **Question (5 pts)**
   (a) Illustrate the block diagram of an analysis-by-synthesis speech coder.
   (b) Illustrate how it is possible to take into account human perception in such coder.

   Solution

   See slide set `MMSP_2.1_Speech_Coding`.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider a memoryless source X, whose p.d.f. is denoted by $p_X(x)$, with variance σ_x^2
   - T F The distortion introduced by lossy coding decreases linearly with rate.
     FALSE. Distortion decreases exponentially with rate.
   - T F $D_U(R) \geq D_{LM}(R)$, where U and LM denote, respectively, uniform and Lloyd-Max quantization.
     TRUE.
   - T F If $p_X(x)$ is uniform, then $D_U(R) = D_{LM}(R)$
     TRUE.
   - T F The rate-distortion function depends only on σ_x^2, and it does not depend on the shape of the p.d.f. $p_X(x)$
     FALSE. $D(R) = \epsilon 2^{-2R}$, where ϵ depends on $p_X(x)$.

   (b) Consider an image coding system.
   - T F Color components are separately encoded in the RGB colorspace
     FALSE. RGB to YCbCr conversion is applied before coding each component separately.
   - T F The coding efficiency increases by increasing the block size.
     TRUE. Increasing the block size it is possible to exploit long range inter-pixel correlation.
   - T F When using threshold quantization it is possible to determine a-priori the number of bits needed to encode an image.
     FALSE.

   (c) Let I_c and I_r denote, respectively, the current and the reference frame used in motion estimation and compensation
   - T F $PSNR(DFD_1(I_c, I_r)) \geq PSNR(DFD_2(I_c, I_r))$, where DFD_1 and DFD_2 denote, respectively, the displaced frame difference computed using full-pixel and half-pixel precision, respectively.
     FALSE.
   - T F $PSNR(DFD(I_c, I_r; B_1)) \geq PSNR(DFD(I_c, I_r; B_2))$, if $B_1 > B2$
     FALSE. Decreasing the block size enables to achieve a better estimate of the local motion, hence $PSNR(DFD(I_c, I_r; B_1)) \leq PSNR(DFD(I_c, I_r; B_2))$.
   - T F The computational complexity of motion estimation is higher than that of motion compensation.
     TRUE.

5. **Question (Matlab - 10pts)**
   You are asked to implement a PCM coder based on a compander-expander.
   - Read the input signal *speech.wav* and show the histogram of the source.
   - Estimate the variance of the source σ_x^2 and fit a Laplacian distribution

$$p(x) = \frac{\lambda}{2} e^{-\lambda|x|} \tag{11}$$

     where $\lambda = \sqrt{2}/\sigma_x$
   - Quantize the signal with a compander-expander based on the following function.

$$y = T[x] = \begin{cases} \frac{1-e^{-\lambda x}}{2} & x \geq 0 \\ \frac{e^{-\lambda x}}{2} & x < 0 \end{cases} \tag{12}$$

- Plot the rate-distortion function (SNR vs. rate) by varying the quantization step size Δ (Hint: estimate the rate based on the entropy of the quantization symbols)

Solution

```
clear all
close all
clc

% read the signal 'offer.wav' and show the histogram of the source
[x, Fs] = audioread('speech.wav');
x = x(x~=0);
info = audioinfo('speech.wav');
nBits = info.BitsPerSample;

l = linspace(-1,1,2^nBits);
h = hist(x,l);
p = h / sum(h);

hfig = figure;
bar(l,p), xlabel('Amplitudes'), ylabel('Distribution'), axis tight

% estimate the variance of the source and fit Laplacian distribution
sigma2 = var(x);
lambda = sqrt(2)/sqrt(sigma2);
p_lap = lambda/2 * exp(-lambda*abs(l));
p_lap = p_lap / sum(p_lap);

figure(hfig)
hold on
plot(l,p_lap,'r')
hold off
legend('Normalized histogram', 'Laplacian distribution')

% compression
y = zeros(length(x),1);
y(x>=0) = 1/2*(1-exp(-lambda*x(x>=0)));
y(x<0) = 1/2*(exp(-lambda*abs(x(x<0)))-1);
range_y = max(y)-min(y);

% plot the rate-distortion function by varying the quantization step
Delta = linspace(0.5,0.0005,16);

SNR = zeros(length(Delta),1);
R = zeros(length(Delta),1);
for ii = 1:length(Delta)
    % quantization
    y_tilde = Delta(ii)*floor(y / Delta(ii)) + Delta(ii)/2;

    % expansion
    x_tilde = zeros(length(y_tilde),1);
    x_tilde(y_tilde >= 0) = 1/lambda*log(1./(1-2*y_tilde(y_tilde >=0)));
    x_tilde(y_tilde < 0) = -1/lambda*log(1./(2*y_tilde(y_tilde <0)+1));

    % estimate SNR
    SNR(ii) = pow2db(var(x)/var(x-x_tilde));

    % estimate rate
    M = (max(y_tilde)-min(y_tilde))/Delta(ii);
    l = linspace(min(y_tilde),max(y_tilde),M);
    d = hist(y_tilde,l);
    p = d / sum(d);
    p = p(p~=0);
    R(ii) = -sum(p .* log2(p));
end

figure
plot(R,SNR,'b'), xlabel('rate [bps]'), ylabel('SNR [dB]')
```